Vrishin Gattu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.68
Gowtham Junior College	Class XII (Telangana State Board of Intermediate Education)	2021	90.00%
Delhi Public School, Nacharam	Class X (Central Board of Secondary Education)	2019	84.50%

PROJECTS

ChalchitraQA - Evidence-Grounded Video-Based Attributed QA System for Indic Languages

Python, IndicConformer, OpenAI Whisper, Qdrant DB, Gemini API, YouTube API

- Built an evidence-grounded QA pipeline for retrieving and answering questions from **YouTube video playlists**, motivated by QA over *Mahabharata pravachanam*.
- Developed an end-to-end pipeline combining **Indic ASR** *[IndicConformer, OpenAI Whisper]***, YouTube transcripts, and confidence-based transcription correction** to improve the reliability of noisy speech-to-text inputs.
- Segmented transcripts into **30-second video chunks** and generated dense embeddings for **vector-database retrieval** *[Qdrant]*.
- Implemented **dense semantic retrieval** to identify relevant video segments and generate answers grounded in retrieved evidence.
- Designed the system to provide **source-attributed answers with cited video chunks**, enabling users to trace responses back to the supporting evidence.
- **Upcoming:** sparse/lexical retrieval, and timestamped video clip merging from citations for more precise attribution.

TransactiWar -- Secure Digital Wallet & Money Transfer Application

PHP, MySQL, Apache, Docker Compose

- Developed **user search** and **peer-to-peer fund transfer** functionality with server-side validation and transactional database operations.
- Implemented **atomic money transfers** using MySQL transactions to maintain balance consistency and prevent partial/duplicate transfers.
- Built **transaction history** and **auditable transfer records**, enabling users to track completed transfers without exposing sensitive information.
- Added validation for **sender/receiver constraints, transfer amounts, and unauthorized operations**, integrating with the project's authentication and authorization layer.
- Collaborated with the security and authentication modules to enforce **prepared SQL statements, CSRF protection, safe output handling, and authorization checks** across transfer workflows.

PROFESSIONAL EXPERIENCE

Cognizant

Programmer Analyst Trainee - Quality Engineering and Assurance

- Underwent technical training in **Java, Selenium, and TestNG**, focusing on automation testing and software quality assurance practices.
- Gained exposure to **automated test case development, test execution, and quality engineering workflows** using industry-standard testing tools.
- Resigned before completion of the training period to pursue an **MTech in Computer Science at IIT Hyderabad**.

SKILLS

Programming Languages: C++, Python, Java, SQL, PHP

Frameworks and Libraries: Scikit-Learn, PyTorch, NumPy, Pandas, Selenium

Tools & Platforms: Git, GitHub, Linux, Docker

RELEVANT COURSES

Advanced Data Structures and Algorithms, Advanced Computer Networks, Foundations of Machine Learning, Linear Optimization, Approximation Algorithms, Deep Learning, Visual Computing, Network Security

POSITIONS OF RESPONSIBILITY

Administrator - Peer Mentoring Committee, Overseeing work of 10 Peer Mentors over 1 semester (during BTech)

Team Lead - Design Team - Naadh '23, Overseeing design for email templates, social media content (during BTech)

SCHOLASTIC ACHIEVEMENTS

Common Admission Test 2024 - **98.28%ile**

3x State Level contestant at Spelling Bee

2x State Level participation in Quiz contests (Byju's, Incredible India)

EXTRACURRICULAR

Hobbies and Interests: Music, Geopolitics, Video Games